

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280477

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

TANAKA; Yasushi et al.

ELECTRONIC APPARATUS, CONTROL METHOD, AND PROGRAM

Abstract

An electronic apparatus includes: an exterior body; a light-emitting unit that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body; and a processor, in which the processor acquires exterior body information that is information on a state of the exterior body, and execute control of adjusting a light emission aspect of the light-emitting unit based on the acquired exterior body information.

Inventors: TANAKA; Yasushi (Saitama, JP), AIDA; Yukari (Saitama, JP), KAYA; Akimasa (Saitama, JP)

Applicant: FUJIFILM Corporation (Tokyo, JP)

Family ID: 94869173

Assignee: FUJIFILM Corporation (Tokyo, JP)

Appl. No.: 19/064664

Filed: February 26, 2025

Foreign Application Priority Data

JP	2024-031550	Mar. 01, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H05B45/10 (20200101); B41J3/36 (20060101); B41J3/46 (20060101); G02B26/02 (20060101); G03B27/00 (20060101); H05B45/20 (20200101); H05B47/11 (20200101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 USC 119 from Japanese Patent Application No. 2024-031550 filed on Mar. 1, 2024, the disclosure of which is incorporated by reference herein.

BACKGROUND

1. Technical Field

[0002] The present disclosed technology relates to an electronic apparatus, a control method, and a program.

2. Related Art

[0003] JP2020-55282A discloses a portable image forming apparatus comprising: a recording unit that records an image on a recording material; an apparatus main body that accommodates the recording unit therein; an attachable/detachable unit that is configured to be attachable to and detachable from the apparatus main body and that is electrically connected to the apparatus main body; a pressurizing unit that presses the attachable/detachable unit in a direction in which a unit-side connection portion of the mounted attachable/detachable unit faces a main body-side connection portion, the direction being orthogonal to an attachment/detachment direction of the attachable/detachable unit; and a movable member that holds the pressurizing unit and that is movable between a first position closest to the main body-side connection portion and a second position farthest from the main body-side connection portion.

[0004] JP3196184Y discloses a battery in which an electronic circuit board in which a charging/discharging module comprising a secondary battery is mounted is accommodated in a housing consisting of a front case and a rear case, the secondary battery is charged by an external power supply, power of the secondary battery can be supplied to a portable electronic apparatus, a decorative LED lamp is mounted on the electronic circuit board, at least any one of the front case or the rear case is configured to allow transmission of light emitted from the decorative LED lamp, and the decorative LED lamp is faced inside at least any one of the front case or the rear case configured to allow the transmission of the light emitted from the decorative LED lamp.

[0005] JP2012-34448A discloses a portable electronic apparatus comprising: a housing; a power generation unit that is disposed on an outer surface of the housing and that generates power via photoelectric conversion based on light incident from the outside; an ultraviolet detection unit that detects illuminance of ultraviolet rays included in the light; a secondary battery that is chargeable with the power generated by the power generation unit; a posture detection unit that is capable of detecting a posture of the housing; a controller that controls a charging state in which the secondary battery is charged with the power generated by the power generation unit; a display unit and an operation unit that are disposed in the housing and that include a light source unit; and a determination unit that determines a chargeable state in accordance with the illuminance of the ultraviolet rays detected by the ultraviolet detection unit and the posture of the housing detected by the posture detection unit, in which the controller performs control of charging the secondary battery and driving the light source unit in a suppressed manner in a case in which the determination unit determines that the state is the chargeable state, and performs control of normally driving the light source unit in a case in which the determination unit determines that the state is not the chargeable state.

[0006] JP2012-156364A discloses a portable electronic apparatus comprising: a falling detection unit that detects that the portable electronic apparatus has fallen into water; a light emitter that is

provided inside a housing; and a light emission control circuit that causes the light emitter to emit light in a case in which the falling detection unit detects the fall into water, in which a translucent portion is provided in a part of the housing.

SUMMARY

[0007] In a case in which a light-emitting unit is provided inside an exterior body of the electronic apparatus, a state in which the light emitted from the inside of the exterior body is visually recognized may be changed from a state intended in advance depending on a state of the exterior body.

[0008] Therefore, the present disclosed technology provides an electronic apparatus, a control method, and a program, which are capable of making a visual recognition state of light transmitted through an exterior body close to a state intended in advance.

[0009] A first aspect according to the present disclosed technology relates to an electronic apparatus comprising: an exterior body; a light-emitting unit that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body; and a processor, in which the processor acquires exterior body information that is information on a state of the exterior body, and executes control of adjusting a light emission aspect of the light-emitting unit based on the acquired exterior body information.

[0010] A second aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the exterior body information includes exterior body color information that is information on a color of the exterior body, and the control includes control of adjusting a light emission color of the light-emitting unit based on the exterior body color information.

[0011] A third aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the exterior body information includes transmitted light color information that is information on a change in a color of the light transmitted through the exterior body, and the control includes control of adjusting a light emission color of the light-emitting unit based on the transmitted light color information.

[0012] A fourth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the exterior body information includes transmitted light intensity information that is information on a change in intensity of the light transmitted through the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting unit based on the transmitted light intensity information.

[0013] A fifth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the processor further acquires ambient light information that is information on ambient light of a periphery of the exterior body, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the acquired ambient light information.

[0014] A sixth aspect according to the present disclosed technology relates to the electronic apparatus according to the fifth aspect, in which the ambient light information includes ambient light intensity information indicating brightness of the periphery of the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting unit based on the ambient light intensity information.

[0015] A seventh aspect according to the present disclosed technology relates to the electronic apparatus according to the fifth aspect, in which the ambient light information includes ambient light color information indicating a color of the ambient light of the periphery of the exterior body, and the control includes control of adjusting a light emission color of the light-emitting unit based on the ambient light color information.

[0016] An eighth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the exterior body information includes information on a state of a movable portion of the exterior body, and the control includes control of adjusting

the light emission aspect of the light-emitting unit based on the information on the state of the movable portion.

[0017] A ninth aspect according to the present disclosed technology relates to the electronic apparatus according to the eighth aspect, in which the movable portion is a cover member that is provided at a position facing the light-emitting unit in the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting unit in accordance with an open/close state of the cover member.

[0018] A tenth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the exterior body information includes posture information indicating a posture of the electronic apparatus, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the posture information.

[0019] An eleventh aspect according to the present disclosed technology relates to the electronic apparatus according to any one of the second to ten aspects, in which the exterior body information includes operation state information indicating an operation state of the electronic apparatus, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the operation state information.

[0020] A twelfth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which a stripe pattern consisting of periodic recessed and protruding portions is formed on a front surface of the exterior body.

[0021] A thirteenth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the exterior body has a display unit that shows a character, a symbol, and/or a figure by using a difference in a three-dimensional shape with a periphery on a front surface, and the display unit is formed at a position facing the light-emitting unit.

[0022] A fourteenth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which a coating film is formed on an outer side of the exterior body, and a thickness of the coating film in a region through which the light from the light-emitting unit is transmitted is set to be less than a thickness of the coating film in another region.

[0023] A fifteenth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the light-emitting unit includes an LED light source, and the control includes control of adjusting a light emission aspect of the LED light source.

[0024] A sixteenth aspect according to the present disclosed technology relates to a control method of an electronic apparatus including an exterior body and a light-emitting unit that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body, the control method comprising: acquiring exterior body information that is information on a state of the exterior body; and executing control of adjusting a light emission aspect of the light-emitting unit based on the acquired exterior body information.

[0025] A seventeenth aspect according to the present disclosed technology relates to a program of controlling an electronic apparatus including an exterior body and a light-emitting unit that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body, the program causing a computer to execute a process comprising: acquiring exterior body information that is information on a state of the exterior body; and executing control of adjusting a light emission aspect of the light-emitting unit based on the acquired exterior body information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] Exemplary embodiments of the technology of the disclosure will be described in detail based on the following figures, wherein:

[0027] FIG. **1** is a conceptual diagram illustrating an example of a configuration of an electronic apparatus according to an embodiment;

[0028] FIG. **2** is a plan view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0029] FIG. **3** is a cross-sectional view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0030] FIG. **4** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0031] FIG. **5** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0032] FIG. **6** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0033] FIG. **7** is a cross-sectional view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0034] FIG. **8** is a side view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0035] FIG. **9** is a block diagram illustrating an example of a hardware configuration of an electrical system of the electronic apparatus according to the embodiment;

[0036] FIG. **10** is a conceptual diagram illustrating an example of main functions of a processor of the electronic apparatus according to the embodiment;

[0037] FIG. **11** is a flowchart illustrating an example of light emission control processing according to the embodiment;

[0038] FIG. **12** is a conceptual diagram illustrating an example of the main functions of the processor of the electronic apparatus according to the embodiment;

[0039] FIG. **13** is a conceptual diagram illustrating an example of the main functions of the processor of the electronic apparatus according to the embodiment;

[0040] FIG. **14** is a conceptual diagram illustrating an example of the main functions of the processor of the electronic apparatus according to the embodiment;

[0041] FIG. **15** is a conceptual diagram illustrating an example of the main functions of the processor of the electronic apparatus according to the embodiment; and

[0042] FIG. **16** is a conceptual diagram illustrating an example of the main functions of the processor of the electronic apparatus according to the embodiment.

DETAILED DESCRIPTION

[0043] An example of an embodiment of an electronic apparatus **10** according to the present disclosed technology will be described with reference to the accompanying drawings.

[0044] Hereinafter, for convenience of description, a width direction, a front-rear direction, and a height direction of the electronic apparatus **10** are indicated by three arrows X, Y, and Z. First, the height direction is indicated by the arrow Z, a direction indicated by the arrow Z is referred to as an up direction of the electronic apparatus **10**, and an opposite direction thereof is referred to as a down direction. The width direction is indicated by the arrow X orthogonal to the arrow Z, a direction indicated by the arrow X is referred to as a right direction of the electronic apparatus **10**, and an opposite direction thereof is referred to as a left direction. The front-rear direction is indicated by the arrow Y in a direction orthogonal to the arrow Z and the arrow X, a direction indicated by the arrow Y is referred to as a front direction of the electronic apparatus **10**, and an opposite direction thereof is referred to as a rear direction. In addition, in the following description, the expressions using the side, such as an upper side, a lower side, a left side, a right side, a front

side, and a rear side, have the same meaning as the expressions using the direction.

First Embodiment

[0045] As illustrated in FIG. 1 as an example, the electronic apparatus **10** has, for example, a printer function. The electronic apparatus **10** is, for example, a portable electronic apparatus. The electronic apparatus **10** is, for example, a mobile printer. Hereinafter, for convenience of description, the embodiment will be described with the mobile printer **11** as an example. The electronic apparatus **10** is an example of an “electronic apparatus” according to the present disclosed technology.

[0046] The mobile printer **11** can communicate with an external device. In the example illustrated in FIG. 1, the mobile printer **11** can communicate with a smartphone **100**. Examples of a communication method include wireless communication (for example, Bluetooth (registered trademark)), but wired communication may be used. The mobile printer **11** receives image data IM transmitted from the smartphone **100** and prints a subject image shown by the received image data IM on a film F. The film F is, for example, a film containing a photosensitive material (so-called instant film).

[0047] The mobile printer **11** comprises an exterior body **12**. In the example illustrated in FIG. 1, the exterior body **12** has a rectangular parallelepiped-shaped housing structure having a longitudinal direction in an X direction. The exterior body **12** accommodates, inside, electronic components and the like for implementing the printer function and other functions of the mobile printer **11**. In addition, for example, the printed film F is discharged from an outlet **14** provided on a left side surface of the exterior body **12**. The exterior body **12** is an example of an “exterior body” according to the present disclosed technology.

[0048] A display unit (display) **16** is formed on the exterior body **12**. The display unit **16** shows at least one of a character, a symbol, or a figure on a front surface of the exterior body **12**. In the example illustrated in FIG. 1, the display unit **16** is formed on an upper surface **12A** of the exterior body **12**. In the example illustrated in FIG. 1, the display unit **16** shows the characters “ABCD”. Examples of the content shown by the display unit **16** include a company name, a product name, a personal name, a catchphrase, a logo mark, and/or a pictogram. In addition, examples of the content shown by the display unit **16** may include a symbol (for example, an arrow indicating a film discharge direction) or a figure (for example, a mark of a power button) for describing an operation of the mobile printer **11**. The display unit **16** is an example of a “display unit” according to the present disclosed technology.

[0049] As illustrated in FIG. 2 as an example, a linear stripe pattern is formed on the exterior body **12** along the longitudinal direction (here, the X direction). Specifically, the stripe pattern consisting of periodic recessed and protruding portions is formed on the front surface of the exterior body **12**. Hereinafter, the stripe pattern is described with the upper surface **12A** on the front surface of the exterior body **12** as an example, but it goes without saying that the stripe pattern may be formed on other surfaces of the exterior body **12** in the same manner.

[0050] In a case of being viewed from a cross section having the longitudinal direction as a normal direction (in a cross-sectional view), protruding portions **18** are periodically formed on the upper surface **12A** of the exterior body **12**. In a cross-sectional view, the protruding portion **18** has a triangular shape, and a portion at a vertex is continuous along the longitudinal direction, whereby the linear stripe pattern is formed. As described above, the protruding portions **18** are periodically present, and thus the stripe pattern consisting of periodic recessed and protruding portions is formed.

[0051] Here, the display unit **16** shows at least one of the character, the symbol, or the figure due to a difference in a three-dimensional shape from a periphery on the front surface of the exterior body **12**. In the example illustrated in FIG. 2, the display unit **16** protrudes to an outer side (here, an upper side) of the exterior body **12** rather than the periphery. As a result, the difference in the three-dimensional shape from the periphery is generated. Specifically, in the above-described stripe

pattern, a portion corresponding to the display unit **16** is the protruding portion **20** that is higher than the protruding portion **18** present on the periphery. That is, in the example illustrated in FIG. 2, the difference in the three-dimensional shape is a difference in a height of the protruding portion of the stripe pattern.

[0052] The difference in the three-dimensional shape between the display unit **16** and the other region causes a difference in a way of reflecting natural light (here, ambient light). In this way, the display unit **16** can be visually recognized due to the difference in the three-dimensional shape from the periphery (here, the characters “ABCD” appear to be floating).

[0053] As illustrated in FIG. 3 as an example, the exterior body **12** includes a first member **22** and a second member **24** disposed adjacent to the first member **22**. The first member **22** is, for example, a cover member that constitutes the upper side of the exterior body **12**, and the second member **24** is, for example, a cover member that constitutes a lower side of the exterior body **12**. The mobile printer **11** comprises a light-emitting unit (light-emitting device) **26** inside the exterior body **12**. The light-emitting unit **26** is provided at a position facing the display unit **16** inside the exterior body **12**. The light-emitting unit **26** can emit emission light L to the outside through the exterior body **12**. The emission light L is, for example, light including a wavelength in a visible light range. The light-emitting unit **26** is an example of a “light-emitting unit” according to the present disclosed technology.

[0054] In the example illustrated in FIG. 3, the light-emitting unit **26** includes, for example, light-emitting diode (LED) light sources **26A** to **26C**. The LED light sources **26A** to **26C** are linearly arranged in an order from a right side inside the exterior body **12**. Here, the form example is described in which the three LED light sources **26A** to **26C** are disposed, but this is merely an example, and two or less or four or more LED light sources may be used. In addition, a disposition interval of the LED light sources, positions at which the LED light sources are disposed, and a disposition method (for example, disposing the LED light sources in a circular shape or disposing the LED light sources at positions of vertices of a quadrangle) can also be set as appropriate. In addition, each of the LED light sources **26A** to **26C** includes RGB light-emitting elements, and a color and a light amount of the emission light L can be changed by controlling a light emission aspect of each light-emitting element. The LED light sources **26A** to **26C** are examples of an “LED light source” according to the present disclosed technology.

[0055] In addition, in the example illustrated in FIG. 3, the light-emitting unit **26** is accommodated inside the first member **22** that constitutes an upper portion of the exterior body **12**. The first member **22** comprises a contact portion **23** as a portion in contact with the second member **24**. Specifically, the first member **22** includes a bottom plate portion **22A** as a plate-shaped portion having a plate thickness direction in the height direction, and a side wall portion **22B** that is erected from a peripheral edge of the bottom plate portion **22A** along the height direction. Here, the side wall portion **22B** is erected toward the second member **24** in a state in which the first member **22** and the second member **24** are combined. Although the details will be described later, the side wall portion **22B** is in contact with the second member **24**, and the side wall portion **22B** also functions as the contact portion **23**.

[0056] The light-emitting unit **26** is provided at a position facing an inner surface of the bottom plate portion **22A**. The emission light L from the light-emitting unit **26** is transmitted through the bottom plate portion **22A** and is emitted to the outside of the exterior body **12**. Here, the transmission of the emission light L includes that the emission light L is scattered in the exterior body **12** and is emitted to the outside. Here, since the exterior body **12** is formed of a resin material, the emission light L is scattered in a case of passing through the bottom plate portion **22A**, and an emission range of the emission light L to the outside is widened.

[0057] In addition, a space **28** in which the film F is accommodatable is formed in the second member **24** constituting a lower portion of the exterior body **12**. The film F is accommodated inside the space **28** via a cover (not illustrated). Here, for convenience of description, one film F is

illustrated, but it goes without saying that a plurality of films F may be accommodated in the space **28**. The space **28** can be open with respect to a lower surface of the exterior body **12**. In other words, the light-emitting unit **26** is provided at a position facing the display unit **16** formed on a surface (here, the upper surface) of the exterior body **12** opposite to a side on which the film F is accommodated. The second member **24** also has the same shape as the first member **22**.

[0058] As illustrated in FIG. 4 as an example, a transmission portion **36**, which is a region through which the emission light L is transmitted through the exterior body **12**, is determined by a distance between the light-emitting unit **26** and the inner surface of the exterior body **12** and a spread angle of the emission light L. Specifically, the transmission portion **36** is determined by the distance between the LED light sources **26A** to **26C** of the light-emitting unit **26** and an inner surface **22A1** of the bottom plate portion **22A** and the spread angle of the emission light L from the LED light sources **26A** to **26C**. In other words, the first member **22** of the exterior body **12** has the transmission portion **36** that allows the transmission of the emission light L from the light-emitting unit **26**. The transmission portion **36** is an example of a “transmission portion” according to the present disclosed technology.

[0059] In the mobile printer **11**, the light-emitting unit **26** may emit light (for example, light up on or blink). Accordingly, it is possible to notify of an operation state of the mobile printer **11** or to improve a user experience (for example, to improve the fun that a user feels during use). Here, in a case in which the exterior body **12** of the mobile printer **11** is viewed in a plan view (here, in a case of being viewed from the upper surface), the light emission via the light-emitting unit **26** overlaps a position of the display unit **16** (that is, the light-emitting unit **26** faces the display unit **16**). In such a case, it is difficult to see the display unit **16** due to the light emission of the light-emitting unit **26**. That is, in a case in which the light-emitting unit **26** emits light, the emission light L transmitted through the exterior body **12** is easily visually recognized, and reflected light (for example, reflected light caused by the difference in three-dimensional shape from the periphery) in the display unit **16** is not easily visually recognized. As a result, the visibility of the display unit **16** is decreased.

[0060] Therefore, as illustrated in FIG. 5 as an example, in the present embodiment, in the exterior body **12**, a difference in transmittance of the emission light L emitted from the light-emitting unit **26** is generated between a region corresponding to the display unit **16** and a region on the periphery. Specifically, a transmission suppression layer **30** containing a material having lower light transmittance than the periphery is formed on the inner surface of the exterior body **12**. More specifically, the transmission suppression layer **30** is formed on the inner surface **22A1** of the bottom plate portion **22A** in the first member **22** of the exterior body **12**. The transmission suppression layer **30** is formed in the region corresponding to the display unit **16**. The region corresponding to the display unit **16** is, for example, a back surface of the display unit **16**, and is a range having the same size as the display unit **16**. In the example illustrated in FIG. 5, transmission suppression layers **30A**, **30B**, **30C**, and **30D** having shapes in which the characters “ABCD” shown by the display unit **16** are inverted are formed. The region corresponding to the display unit **16** need not be a range having the same size as the display unit **16**, and the size may be changed as appropriate.

[0061] The transmission suppression layers **30A**, **30B**, **30C**, and **30D** are formed on the inner surface **22A1** of the first member **22** by printing. Specifically, the transmission suppression layers **30A**, **30B**, **30C**, and **30D** are formed by printing using a coating material containing a material that absorbs the wavelength in the visible light range. For example, the transmission suppression layers **30A**, **30B**, **30C**, and **30D** are printed using a black coating material. A part of the emission light L is absorbed or reflected due to the interaction with the transmission suppression layer **30**. In this way, the transmission of the emission light L in the region corresponding to the display unit **16** is suppressed.

[0062] As described above, the light-emitting unit **26** is provided at a position facing the display

unit **16**. In addition, inside the exterior body **12**, the transmission suppression layer **30** is formed in the region corresponding to the display unit **16**. Therefore, the transmission of a part of the emission light **L** from the light-emitting unit **26** is suppressed by the transmission suppression layer **30**. Here, a part of the emission light **L** is absorbed by the black coating material included in the transmission suppression layer **30**. In addition, in the example illustrated in FIG. 5, in a case in which the first member **22** is viewed from the inner surface side (here, in a case of being viewed from below), the transmission suppression layers **30A**, **30B**, **30C**, and **30D** have the shapes in which the characters “ABCD” are inverted. Therefore, in the region having the shapes in which the characters of “ABCD” are inverted, the transmission of the emission light **L** through the exterior body **12** is suppressed.

[0063] As illustrated in FIG. 6 as an example, in the mobile printer **11**, in a case in which the light-emitting unit **26** emits light, as described above, the transmission of the emission light **L** is suppressed by the transmission suppression layer **30** provided in the region corresponding to the display unit **16**. As a result, the emission light **L** is less likely to be transmitted through the display unit **16**, and thus the display unit **16** is easily visually recognized. In the example illustrated in FIG. 6, the characters “ABCD” of the display unit **16** are easily visually recognized by the transmission suppression layer **30** having the shapes in which the characters “ABCD” are inverted.

[0064] As a result, even in a case in which the region corresponding to the display unit **16** is identified from other portions due to a difference in a degree of visual recognition and the light-emitting unit **26** provided inside the exterior body **12** is lighted up, it is easy to recognize the display unit **16**. In addition, in a range illuminated with the emission light **L**, the content (here, the characters “ABCD”) indicated by the transmission suppression layer **30** floats as a shadow, and the designability is improved.

[0065] A portion of the exterior body **12** facing the light-emitting unit **26** is required to contain a material having relatively high transmittance in order to transmit the emission light **L** from the light-emitting unit **26**. On the other hand, a portion opposite to the portion facing the light-emitting unit **26** is required to contain a material having relatively low transmittance. This configuration is to suppress the entry of unnecessary external light into the inside of the exterior body **12**.

[0066] In a case in which the exterior body **12** is formed by combining two types of members having different optical characteristics (for example, light transmittance), a difference in the optical characteristics makes it noticeable that the two types of members are separate components in a portion in which the two types of members are in contact with each other. As a result, the exterior body **12** may lack a sense of unity as a whole and may impair the exterior characteristics.

[0067] Therefore, as illustrated in FIG. 7 as an example, a thickness of a coating film **32** is adjusted in the first member **22** of the exterior body **12**. The first member **22** includes the coating film **32** and a substrate **34**. The coating film **32** is formed on a front surface of the substrate **34**. The substrate **34** is formed of a translucent material. Here, the translucent material is a material that allows the transmission of the light while scattering the light. The substrate **34** is, for example, a milky white resin. In addition, the coating film **32** is a film formed by fixing the coating material to the front surface of the substrate **34**, and the type of the coating material is set as appropriate in accordance with a color variation of the exterior body **12**. As described above, the emission light **L** from the light-emitting unit **26** is emitted to the outside through the first member **22**. Since the first member **22** is formed of the translucent material, the emission light **L** is easily emitted to the outside of the exterior body **12**, and the emission light **L** is easily visually recognized. The coating film **32** is an example of a “coating film” according to the present disclosed technology.

[0068] The coating film **32** has a thickness t_1 in the side wall portion **22B** of the first member **22**. On the other hand, the coating film **32** has a thickness t_2 in the bottom plate portion **22A** of the first member **22**. In addition, the thickness t_1 of the coating film **32** in the side wall portion **22B** is greater than the thickness t_2 of the coating film **32** in the bottom plate portion **22A**. In other words, in the transmission portion **36** (see FIG. 4) through which the emission light **L** is transmitted in the

bottom plate portion **22A**, the thickness **t2** of the coating film **32** is less than the thickness **t1** of the coating film **32** in the side wall portion **22B**.

[0069] Specifically, the thickness **t1** of the coating film **32** in the side wall portion **22B** is 1.5 times or more the thickness **t2** of the coating film **32** in the bottom plate portion **22A**. More specifically, the thickness **t1** is, for example, about 20 μm , and the thickness **t2** is, for example, about 10 to 12 μm .

[0070] The second member **24** includes a bottom plate portion **24A** and a side wall portion **24B** that is erected from a peripheral edge of the bottom plate portion **24A** along the height direction. The second member **24** is formed of a non-transparent material. Here, the non-transparent material is a material that cuts light. Specifically, the non-transparent material is, for example, a material in which titanium oxide particles are dispersed in a resin as a base material. That is, the second member **24** is formed of, for example, a resin containing titanium oxide. As described above, the space **28** in which the film **F** is accommodated is formed in the second member **24**. Since the second member **24** is formed of the non-transparent material, the entry of the external light into the inside of the second member **24** is suppressed.

[0071] The first member **22** and the second member **24** are joined in a state of being adjacent to each other. Further, the first member **22** is in contact with the second member **24** via the contact portion **23**. In the example illustrated in FIG. 7, the side wall portion **22B** of the first member **22** and the side wall portion **24B** of the second member **24** are in contact with each other. That is, in the first member **22**, the side wall portion **22B** is the contact portion **23**. Specifically, the lower surface of the side wall portion **22B** of the first member **22** and the upper surface of the side wall portion **24B** of the second member **24** are in contact with each other. Then, as described above, the thickness **t1** of the coating film **32** in the side wall portion **22B** is greater than the thickness **t2** of the coating film **32** in the bottom plate portion **22A**. That is, the thickness **t1** of the coating film **32** in the contact portion **23** is greater than the thickness **t2** of the coating film **32** in the transmission portion **36**.

[0072] The first member **22** and the second member **24** are joined to each other in a state in which the side wall portion **22B** and the side wall portion **24B** are in contact with each other. The method of joining the first member **22** and the second member **24** is not particularly limited, and may be, for example, mechanical fastening in which a fixing unit for fitting a claw for fixing into a hole and screwing are combined, or may be joined by an adhesive, ultrasonic welding, or the like.

[0073] As an example, a case is considered in which the thickness **t1** of the coating film **32** in the side wall portion **22B** of the first member **22** (hereinafter, simply referred to as a coating film thickness **t1**) is less than the thickness **t2** of the coating film **32** in the bottom plate portion **22A** of the first member **22** (hereinafter, simply referred to as a coating film thickness **t2**), as illustrated on the left side of FIG. 8. In such a case, in a portion (here, the side wall portion **22B** and the side wall portion **24B**) in which the first member **22** and the second member **24** are in contact with each other, the difference in the light transmittance makes it noticeable that the first member **22** and the second member **24** are separate components.

[0074] On the other hand, as illustrated on the right side of FIG. 8, a case is considered in which the coating film thickness **t1** in the side wall portion **22B** of the first member **22** is greater than the coating film thickness **t2** in the bottom plate portion **22A** of the first member **22**. In such a case, the light transmittance of the side wall portion **22B** of the first member **22** is close to the light transmittance of the side wall portion **24B** of the second member **24**, so that a difference between the first member **22** and the second member **24** is less noticeable. As a result, a sense of unity of the entire exterior body **12** as a whole is increased, and the exterior characteristics are improved.

[0075] Next, a hardware configuration of an electrical system of the mobile printer **11** according to the present embodiment will be described. As illustrated in FIG. 9 as an example, the mobile printer **11** comprises a control device **40**. The control device **40** controls an overall operation of the mobile printer **11**. The control device **40** comprises a processor **42**, a storage **44**, a random-access

memory (RAM) **46**, and an external interface (I/F) **48**. The processor **42**, the RAM **46**, the storage **44**, and the external I/F **48** are connected to a bus **49**. The processor **42** is an example of a “processor” according to the present disclosed technology. The control device **40** is an example of a “computer” according to the present disclosed technology.

[0076] A memory is connected to the processor **42**. The memory includes the storage **44** and the RAM **46**. The processor **42** has, for example, a central processing unit (CPU). It should be noted that the processor **42** may be provided with a graphics processing unit (GPU) dedicated to image processing, separately from the CPU.

[0077] The RAM **46** is a memory that stores the information and is used as a work memory by the processor **42**. Examples of the RAM **46** include a dynamic random access memory (DRAM) or a static random access memory (SRAM).

[0078] The storage **44** is a non-volatile storage device that stores various programs, various parameters, and the like. Examples of the storage **44** include a flash memory (for example, an electrically erasable and programmable read only memory (EEPROM) and a solid state drive (SSD)), and/or a hard disk drive (HDD). It should be noted that the flash memory and the HDD are merely examples, and at least one of the flash memory, the HDD, a magnetoresistive memory, or a ferroelectric memory may be used as the storage **44**.

[0079] The external I/F **48** is responsible for exchanging various types of information with devices present outside the control device **40**. The external I/F **48** is communicably connected to the light-emitting unit **26**, a reception device **38**, and a detection unit **50**.

[0080] The light-emitting unit **26** can emit the light L having various light emission colors and light emission intensities in accordance with the setting of the parameters. Specifically, as described above, the mobile printer **11** comprises the LED light source **26A**, the LED light source **26B**, and the LED light source **26C** as the light-emitting units **26**. The LED light source **26A** includes a red (R) light-emitting element **27A**, a green (G) light-emitting element **27B**, and a blue (B) light-emitting element **27C**. By changing the setting of the parameters related to the light emission of each light-emitting element of the LED light source **26A**, the light amount and the light emission color of the LED light source **26A** are adjusted.

[0081] Similarly, the LED light sources **26B** and **26C** also includes the red (R) light-emitting element **27A**, the green (G) light-emitting element **27B**, and the blue (B) light-emitting element **27C**. It should be noted that, in the following description, in a case in which it is not necessary to distinguish between the red (R) light-emitting element **27A**, the green (G) light-emitting element **27B**, and the blue (B) light-emitting element **27C**, the red (R) light-emitting element **27A**, the green (G) light-emitting element **27B**, and the blue (B) light-emitting element **27C** may be simply referred to as a “light-emitting element **27**”.

[0082] In addition, in the LED light sources **26B** and **26C**, the light amount and the light emission color of the LED light sources **26B** and **26C** are also adjusted by changing the setting of the parameters of the light-emitting element **27**. As a result of adjusting the light amounts and the light emission colors of the LED light source **26A** to the LED light source **26C**, the emission light L is emitted from the light-emitting unit **26**.

[0083] The light-emitting element **27** is an LED that emits a pulse of light in accordance with a periodically input driving pulse. Examples of the parameters of the light-emitting element **27** include a duty ratio of the driving pulse, and/or an emission time determined by the number of repetitions of the driving pulse.

[0084] The detection unit **50** can detect information on the exterior body **12** of the mobile printer **11**. In the example illustrated in FIG. **9**, an optical sensor **51** is illustrated as the detection unit **50**. The optical sensor **51** is, for example, a sensor that can detect the color and the illuminance of the light incident on the light-receiving element.

[0085] As described above, the light-emitting unit **26** is caused to emit light, and it is possible to notify of the operation state of the mobile printer **11** or to improve the user experience. In such a

case, the light L emitted from the light-emitting unit **26** is affected by the state of the exterior body **12**. For example, in a case in which the exterior body **12** presents a red color, it is assumed that white light is emitted from the light-emitting unit **26**. In such a case, even though the white light is intended to be visually recognized, the emission light L is transmitted through the red exterior body **12**, so that the emission light L having a reddish color is visually recognized by the user. As described above, a visual recognition state of the emission light L by the user is changed from the intended state. Here, the visual recognition state of the user includes the color of the emission light L recognized by the user and a degree (for example, brightness) of intensity of the emission light L recognized by the user.

[0086] Therefore, in view of such circumstances, in the mobile printer **11** according to the present embodiment, in the control device **40**, the processor **42** reads out a control program **44A** from the storage **44** and executes the readout control program **44A** on the RAM **46**. As a result, the processor **42** operates as an acquisition unit **42A** and a light source controller **42B**. The control program **44A** is an example of a “program” according to the present disclosed technology.

[0087] As illustrated in FIG. **10** as an example, in the processor **42**, the acquisition unit **42A** acquires exterior body information **52** from the storage **44**. The exterior body information **52** is information on the exterior body **12**. The exterior body information **52** includes information on the exterior body **12** itself, and includes information on the periphery of the exterior body **12**. The exterior body information **52** includes exterior body color information **54**. The exterior body color information **54** is information on the color of the exterior body **12**. The exterior body color information **54** is stored in the storage **44** in, for example, a manufacturing stage of the mobile printer **11**. In addition, for example, the exterior body color information **54** may be obtained by detecting the color of the exterior body **12** via the detection unit **50**. Further, in a case in which the exterior body **12** is replaced, the exterior body color information **54** may be updated by reading a contactless storage medium (not illustrated) provided in the exterior body **12** with a contactless reading device (not illustrated). The acquisition unit **42A** outputs the exterior body color information **54** to the light source controller **42B**. The exterior body information **52** is an example of “exterior body information” according to the present disclosed technology, and the exterior body color information **54** is an example of “exterior body color information” according to the present disclosed technology.

[0088] The light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** based on the exterior body information **52**. Specifically, the light source controller **42B** acquires the exterior body color information **54** from the acquisition unit **42A**. In addition, the light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** in accordance with the color of the exterior body **12** indicated by the exterior body color information **54**. Here, the control of the light emission aspect includes control of the light emission color of the light-emitting unit **26** and control of the light emission amount of the light-emitting unit **26**. In addition, the control of the light emission aspect also includes control of whether or not the light emission of the light-emitting unit **26** (for example, light up or blink) is performed, or control of which of the plurality of LED light sources **26A** to **26C** emits light or in what order the light is emitted.

[0089] Specifically, the light source controller **42B** executes control of setting the parameters. More specifically, the light source controller **42B** acquires a setting table **56** from the storage **44**. The setting table **56** is a table in which the color of the exterior body **12** is set as input information and the parameters of the light emission conditions of the light-emitting unit **26** are set as output information. The setting table **56** is obtained, for example, by adjusting the light emission color in accordance with the color of the exterior body **12** in a sensory test using an actual machine and setting the light L transmitted through the exterior body to a reference color.

[0090] In the example illustrated in FIG. **10**, in a case in which red or blue is the input information as the color of the exterior body **12** in the setting table **56**, the parameters of the respective LED

light sources **26A** to **26C** are illustrated as the corresponding output information. As the parameters, the duty ratio (ratio between a pulse width and a pulse interval) of the driving pulse of each of the light-emitting elements **27** of red (R), green (G), and blue (B) constituting each of the LED light sources **26A** to **26C** is illustrated, and a total light emission time T of each of the LED light sources **26A** to **26C** is illustrated. It should be noted that, although not illustrated, the parameters are also held in the setting table **56** for the color of the exterior body **12** other than red and blue.

[0091] The light source controller **42B** reads out the parameters of the LED light sources **26A** to **26C** based on the exterior body color information **54** by using the setting table **56**. Specifically, the parameters of the LED light sources **26A** to **26C** corresponding to the color of the exterior body **12** indicated by the exterior body color information **54** are read out. Then, the light source controller **42B** executes the setting of the readout parameters of the LED light sources **26A** to **26C**.

Specifically, the duty ratios of the driving pulses of the light-emitting elements **27** of red (R), green (G), and blue (B) constituting the respective LED light sources **26A** to **26C** are set, and the total light emission time T of the respective LED light sources **26A** to **26C** is set. As described above, the light source controller **42B** executes the control of adjusting the light emission aspects of the LED light sources **26A** to **26C**.

[0092] Adjusting the duty ratio of the driving pulse of the light-emitting element **27** of each color means adjusting the light emission amount per unit time of the light-emitting element **27** of each color. Then, by a combination of the light emission amount of the light-emitting elements **27** of each color, it is possible to produce light of various colors as the LED light sources **26A** to **26C** which are a set of the light-emitting elements **27** of each color.

[0093] For example, in a case in which the light emission amount of the light-emitting element **27** of red (R) is relatively large, the light of the LED light sources **26A** to **26C** is increased in redness, and in a case in which the light emission amount of the light-emitting element **27** of blue (B) is relatively large, the light of the LED light sources **26A** to **26C** is increased in blueness. By making the light emission amount of each light-emitting element **27** uniform, the light from the LED light sources **26A** to **26C** is the white light. In addition, by finely adjusting the light emission amount of the light-emitting element **27** of each color, it is possible to produce the light of a wider variety of colors. Further, in a case in which each of the LED light sources **26A** to **26C** is caused to emit light of different colors, it is also possible to change the color for each region of each of the LED light sources **26A** to **26C**. As described above, adjusting the light amount ratio of the light emitted from the light-emitting elements **27** of red (R), green (G), and blue (B) in the light emission colors of the LED light sources **26A** to **26C** and changing the light emission colors in the light emission colors of the LED light sources **26A** to **26C** make it possible to emit the emission light L corresponding to the color of the exterior body **12**.

[0094] For example, in a case in which the color of the exterior body **12** is red, and the white light is desired to be emitted (that is, in a case in which the intended light emission color is white), the light emission amount of the light-emitting element **27** of red (R) is relatively reduced to make the color of the emission light L transmitted through the exterior body **12** close to white. In addition, for example, in a case in which the color of the exterior body **12** is blue, and the white light is desired to be emitted (that is, in a case in which the intended light emission color is white), the light emission amount of the light-emitting element **27** of blue (B) is relatively reduced to make the color of the emission light L transmitted through the exterior body **12** close to white.

[0095] Here, although the form example is described in which the parameters are set by using the setting table **56**, this is merely an example. Instead of the setting table **56**, a parameter derivation expression (not illustrated) representing the setting table **56** as a function may be used to set the parameters. The parameter derivation expression is an arithmetic expression in which a value indicating the exterior body color is an independent variable and a parameter value is a dependent variable.

[0096] The light source controller **42B** determines whether or not a light emission timing of the

light-emitting unit **26** has arrived. Examples of the light emission timing of the light-emitting unit **26** include a case in which the light-emitting unit **26** emits light while the mobile printer **11** is being activated, while the mobile printer **11** is communicating with the external device, or while the mobile printer **11** is performing a printing operation on the film **F**. In a case in which the light emission timing of the light-emitting unit **26** has arrived, the light source controller **42B** causes the LED light sources **26A** to **26C** to emit light. As described above, the light emission aspects of the LED light sources **26A** to **26C** are adjusted in accordance with the color of the exterior body **12**. The light **L** emitted from the light-emitting unit **26** is transmitted through the exterior body **12**. In a case in which the light **L** transmitted through the exterior body **12** is recognized through eyes **E** of the user, the light **L** is recognized in a state close to the light emission color intended in advance. [0097] Next, light emission control processing in the mobile printer **11** according to the present embodiment will be described with reference to FIG. **11**. FIG. **11** is a flowchart illustrating an example of the light emission control processing. A flow of the light emission control processing illustrated in FIG. **11** is an example of a “control method” according to the present disclosed technology.

[0098] In the light emission control processing illustrated in FIG. **11** as an example, first, in step **ST12**, the acquisition unit **42A** acquires the exterior body information **52** from the storage **44**. The exterior body information **52** includes the exterior body color information **54**. After the processing in step **ST12** is executed, the light emission control processing proceeds to step **ST14**.

[0099] In step **ST14**, the light source controller **42B** controls the light emission aspect of the light-emitting unit **26** in accordance with the exterior body information **52** acquired in step **ST12**. Specifically, the light source controller **42B** acquires the setting table **56** from the storage **44** to set the parameters of each of the LED light sources **26A** to **26C** by using the acquired setting table **56**. After the processing in step **ST14** is executed, the light emission control processing proceeds to step **ST16**.

[0100] In step **ST16**, the light source controller **42B** determines whether or not the light emission timing of the light-emitting unit **26** has arrived. In step **ST16**, in a case in which the light emission timing has arrived, a Yes determination is made, and the light emission control processing proceeds to step **ST18**. In step **ST16**, in a case in which the light emission timing has not arrived, a No determination is made, and the light emission control processing proceeds to step **ST16** again.

[0101] In step **ST18**, the light source controller **42B** causes the light-emitting unit **26** to emit light. After the processing in step **ST18** is executed, the light emission control processing proceeds to step **ST20**.

[0102] In step **ST20**, the light source controller **42B** determines whether or not a condition (hereinafter, referred to as an “end condition”) for the end of the light emission control processing is satisfied. Examples of the end condition include a condition in which an instruction to end the light emission control processing is received by the reception device **38**. In step **ST20**, in a case in which the end condition is not satisfied, a No determination is made, and the light emission control processing proceeds to step **ST12**. In step **ST20**, in a case in which the end condition is satisfied, a Yes determination is made, and the light emission control processing ends.

[0103] As described so far, the electronic apparatus **10** according to the first embodiment includes the exterior body **12** and the light-emitting unit **26**. The light-emitting unit **26** is provided inside the exterior body **12**. In addition, the light-emitting unit **26** can emit the light **L** to the outside through the exterior body **12**. Further, the electronic apparatus **10** comprises the processor **42**, and, in the processor **42**, the acquisition unit **42A** acquires the exterior body information **52**. The light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** based on the acquired exterior body information **52**. Accordingly, since the light emission aspect of the light-emitting unit **26** is adjusted in accordance with the state of the exterior body **12**, the visual recognition state of the light **L** transmitted through the exterior body **12** is easily made to be close to the state intended in advance, as compared to a case in which the visual recognition state of the

light L transmitted through the exterior body **12** is always the same light emission aspect even in a case in which the state of the exterior body **12** is changed.

[0104] In addition, in the electronic apparatus **10** according to the first embodiment, the exterior body information **52** includes the exterior body color information **54** that is information on the color of the exterior body **12**. The control executed by the light source controller **42B** includes control of adjusting the light emission color of the light-emitting unit **26** based on the exterior body color information **54**. Accordingly, since the light emission color of the light-emitting unit **26** is adjusted in accordance with the color of the exterior body **12**, the visually recognized color of the light L transmitted through the exterior body **12** is easily made to be close to a state intended in advance as compared to a case in which the light emission color is always the same even in a case in which the color of the exterior body **12** is changed.

[0105] In addition, in the electronic apparatus **10** according to the first embodiment, the stripe pattern consisting of the periodic recessed and protruding portions is formed on the front surface of the exterior body **12**, and the difference in the three-dimensional shape in the display unit **16** is the difference in the height of the protruding portion of the stripe pattern. In a case in which the stripe pattern is formed on the front surface of the exterior body **12** in order to enhance the designability, the display unit **16** is formed by the difference in the height of the three-dimensional shape. In such a case, the emission light L is likely to be transmitted from the valleys of the recessed and protruding portions of the stripe pattern, but, in the present configuration, since the light emission aspect of the light-emitting unit **26** can be adjusted, the light L transmitted through the exterior body **12** is likely to be recognized.

[0106] In the electronic apparatus **10** according to the first embodiment, the display unit **16** is formed on the exterior body **12**. The display unit **16** shows at least one of the character, the symbol, or the figure by the difference in the three-dimensional shape from the periphery on the front surface of the exterior body **12**. Further, the display unit **16** is provided at a position facing the light-emitting unit **26**. In a case in which the display unit **16** indicating the character or the like is formed by the difference in the three-dimensional shape, the difference in the transmittance of the light L is likely to be generated in the exterior body **12**. In the present configuration, since the light emission aspect of the light-emitting unit **26** can be adjusted, it is easy to recognize the light L transmitted through the exterior body **12**.

[0107] In addition, in the electronic apparatus **10** according to the first embodiment, in the exterior body **12**, the coating film **32** is formed on the outside, and the thickness t_1 of the coating film **32** in the transmission portion **36** through which the light from the light-emitting unit **26** is transmitted is set to be less than the thickness t_2 of the coating film **32** in another region. Even in a case in which the thickness t_2 of the coating film **32** is small in order to easily transmit the light L emitted from the light-emitting unit **26**, in the present configuration, the light L transmitted through the exterior body **12** is easily recognized.

[0108] In addition, in the electronic apparatus **10** according to the first embodiment, the light-emitting unit **26** includes the LED light sources **26A** to **26C**, and the control via the light source controller **42B** includes control of adjusting the light emission aspects of the LED light sources **26A** to **26C**. As a result, since the light emission aspect is adjusted by adjusting the duty ratio of the driving pulse of the LED and the light emission time determined by the number of repetitions of the driving pulse, it is easy to adjust the light emission aspects of the LED light sources **26A** to **26C**.

First Modification Example

[0109] In the first embodiment, although the form example is described in which the light emission aspect of the light-emitting unit **26** is adjusted in accordance with the exterior body color information **54**, the present disclosed technology is not limited to this. In the first modification example, the light emission aspect of the light-emitting unit **26** is adjusted in accordance with transmitted light color information **58** and transmitted light intensity information **60**.

[0110] As illustrated in FIG. 12 as an example, in the processor 42, the acquisition unit 42A acquires the exterior body information 52 from the storage 44. In the first modification example, the exterior body information 52 includes the transmitted light color information 58 and the transmitted light intensity information 60. The transmitted light color information 58 is information on a change in the color of the light L transmitted through the exterior body 12. Specifically, the transmitted light color information 58 is information for specifying a degree of change in the color (hereinafter, also simply referred to as a “color change rate”) of the light L transmitted through the exterior body 12 with respect to the color of the light L emitted from the light-emitting unit 26.

[0111] In addition, the transmitted light intensity information 60 is information on a change in the intensity of the light L transmitted through the exterior body 12. Specifically, the transmitted light intensity information 60 is information for specifying a degree of change in the intensity (for example, brightness) (hereinafter, also simply referred to as an “intensity change rate”) of the light L transmitted through the exterior body 12 with respect to the light L emitted from the light-emitting unit 26. The color change rate and the intensity change rate are affected by, for example, the material of the exterior body 12, the type of the coating material, the thickness of the coating film 32, the front surface shape of the exterior body 12, and the like. The transmitted light color information 58 is an example of “transmitted light color information” according to the present disclosed technology, and the transmitted light intensity information 60 is an example of “transmitted light intensity information” according to the present disclosed technology.

[0112] The transmitted light color information 58 and the transmitted light intensity information 60 are stored in the storage 44, for example, in the manufacturing stage of the mobile printer 11. In addition, the transmitted light color information 58 and the transmitted light intensity information 60 are obtained, for example, by measuring the color and the brightness of the transmitted light via a color and brightness meter in a test using an actual machine and obtaining a rate of change with respect to the intended light emission color and brightness. Then, the acquisition unit 42A outputs the transmitted light color information 58 and the transmitted light intensity information 60 to the light source controller 42B.

[0113] The light source controller 42B acquires the transmitted light color information 58 and the transmitted light intensity information 60 from the acquisition unit 42A. In addition, the light source controller 42B executes control of adjusting the light emission aspect of the light-emitting unit 26 in accordance with the color change rate indicated by the transmitted light color information 58 and the intensity change rate indicated by the transmitted light intensity information 60.

[0114] Specifically, the light source controller 42B executes the control of setting the parameters. More specifically, the light source controller 42B acquires the light emission aspect derivation expression 62 from the storage 44. The light emission aspect derivation expression 62 is an arithmetic expression in which the color change rate and the intensity change rate are independent variables and the parameter value (for example, the duty ratio of the driving pulse or the total light emission time T) is a dependent variable. The light emission aspect derivation expression 62 is obtained, for example, by adjusting the light emission color and the light emission amount in accordance with the measurement result of the transmitted light in a sensory test using an actual machine and matching the light emission color and the brightness with the intended light emission color and brightness.

[0115] For example, in a case in which the red component tends to decrease in the color change rate of the exterior body 12, and the white light is desired to be emitted (that is, in a case in which the intended light emission color is white), the light emission amount of the light-emitting element 27 of red (R) is relatively increased to make the color of the emission light L transmitted through the exterior body 12 close to white. In addition, for example, in a case in which the intensity of the light L tends to decrease by about 10% in the intensity change rate of the exterior body 12, the light emission amount of each light-emitting element 27 is increased by about 10% to bring the

brightness of the emission light L transmitted through the exterior body **12** close to an intended value.

[0116] It should be noted that, here, although the form example is described in which the parameters are set by using the light emission aspect derivation expression **62**, this is merely an example, and a derivation table may be used. The derivation table is a table in which the color change rate and the intensity change rate are input information and the value of the parameter is output information.

[0117] In a case in which the light emission timing of the light-emitting unit **26** has arrived, the light source controller **42B** causes the LED light sources **26A** to **26C** to emit light. As described above, the light emission aspects of the respective LED light sources **26A** to **26C** are adjusted in accordance with the color change rate and the intensity change rate. In a case in which the light L transmitted through the exterior body **12** is recognized through the eyes E of the user, the light L is recognized in a state close to the light emission aspect that is intended in advance.

[0118] As described so far, in the electronic apparatus **10** according to the first modification example, the exterior body information **52** includes the transmitted light color information **58**. The control via the light source controller **42B** includes control of adjusting the light emission color of the light-emitting unit **26** based on the transmitted light color information **58**. Accordingly, since the light emission color is adjusted in accordance with the color change rate of the light L transmitted through the exterior body **12**, the color of the light L transmitted through the exterior body **12** can be made to be close to an intended color as compared to a case in which the light emission color is always the same. For example, in a case in which the exterior body **12** has a color variation, and the appearance of the color of the light L is changed in accordance with the color of the exterior body **12**, it is necessary to change the description of the color of the light L for each color variation. In the present configuration, since the color of the light L transmitted through the exterior body **12** is made to be close to the intended color, the complexity of the description is reduced. In addition, the variation in the appearance of the light emission color of the product is reduced, which can also contribute to the improvement of the quality.

[0119] In the electronic apparatus **10** according to the first modification example, the exterior body information **52** includes the transmitted light intensity information **60**. The control via the light source controller **42B** includes control of adjusting the light emission amount of the light-emitting unit **26** based on the transmitted light intensity information **60**. Accordingly, since the light emission amount is adjusted in accordance with the intensity change rate of the light L transmitted through the exterior body **12**, the intensity (for example, the brightness) of the light L transmitted through the exterior body **12** can be made to be close to the intended intensity of the light as compared to a case in which the light emission amount is always the same.

[0120] It should be noted that, in the first modification example, the form example is described in which the light emission aspect of the light-emitting unit **26** is adjusted based on both the transmitted light color information **58** and the transmitted light intensity information **60**, but the present disclosed technology is not limited to this. For example, an aspect may be adopted in which only the adjustment of the light emission color of the light-emitting unit **26** is performed based on the transmitted light color information **58**, or an aspect may be adopted in which only the adjustment of the light emission amount of the light-emitting unit **26** is performed based on the transmitted light intensity information **60**.

Second Embodiment

[0121] In the first embodiment, although the form example is described in which the light emission aspect is adjusted in accordance with the exterior body information **52**, the present disclosed technology is not limited to this. In the second embodiment, the light emission aspect of the light-emitting unit **26** is adjusted based on the ambient light information **64**.

[0122] As illustrated in FIG. **13** as an example, the mobile printer **11** comprises the optical sensor **51**. The optical sensor **51** detects the ambient light (that is, the external light) on the periphery of

the exterior body **12** via the light-receiving element. Then, the optical sensor **51** outputs the ambient light information **64** to the processor **42** based on a detection result. The ambient light information **64** is information on the ambient light on the periphery of the exterior body **12**. The ambient light information **64** is an example of “ambient light information” according to the present disclosed technology. The ambient light information **64** includes ambient light color information **66** and ambient light intensity information **68**. The ambient light color information **66** is information indicating the color of the ambient light. The ambient light intensity information **68** is information indicating the brightness (that is, the illuminance of the ambient light) of the periphery of the exterior body **12**.

[0123] In the processor **42**, the acquisition unit **42A** acquires the ambient light color information **66** and the ambient light intensity information **68**, as the ambient light information **64**, from the optical sensor **51**. Then, the acquisition unit **42A** outputs the ambient light color information **66** and the ambient light intensity information **68** to the light source controller **42B**.

[0124] The light source controller **42B** acquires the ambient light color information **66** and the ambient light intensity information **68** from the acquisition unit **42A**. The light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** based on the ambient light information **64**. Specifically, the light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** in accordance with the color of the ambient light indicated by the ambient light color information **66** and the brightness of the periphery indicated by the ambient light intensity information **68**.

[0125] Specifically, the light source controller **42B** executes the control of setting the parameters. More specifically, the light source controller **42B** acquires the light emission aspect derivation expression **70** from the storage **44**. The light emission aspect derivation expression **70** is an arithmetic expression in which the color and the illuminance of the ambient light are independent variables and a parameter value (for example, the duty ratio of the driving pulse or the total light emission time T) is a dependent variable. The light emission aspect derivation expression **70** is obtained, for example, by adjusting the light emission color and the light emission amount in accordance with the measurement result of the ambient light in a sensory test using an actual machine.

[0126] For example, in a case in which the red component exceeds a threshold value in the color of the ambient light (for example, indoors with red illumination or outdoors under a sunset), and the white light is desired to be emitted (that is, in a case in which an intended light emission color is white), the light emission amount of the light-emitting element **27** of red (R) is relatively increased. As a result, it is possible to make the color of the emission light L transmitted through the exterior body **12** close to white even in the reddish ambient light. In addition, for example, in a case in which the brightness of the ambient light exceeds a threshold value, the light emission amount from each light-emitting element **27** is increased to bring the light amount of the emission light L transmitted through the exterior body **12** close to an intended value. As a result, the light emission of the light-emitting unit **26** is easily visually recognized even in a dazzling environment, such as under sunlight.

[0127] It should be noted that, here, although the form example is described in which the parameters are set by using the light emission aspect derivation expression **70**, this is merely an example, and a derivation table may be used. The derivation table is a table in which the color and the brightness of the ambient light are input information and the parameter value is output information.

[0128] In a case in which the light emission timing of the light-emitting unit **26** has arrived, the light source controller **42B** causes the LED light sources **26A** to **26C** to emit light. As described above, the light emission aspects of the LED light sources **26A** to **26C** are adjusted in accordance with the color and the brightness of the ambient light. In a case in which the light L transmitted through the exterior body **12** is recognized through the eyes E of the user, the light L is recognized

in a state close to the light emission aspect that is intended in advance.

[0129] As described so far, in the electronic apparatus **10** according to the second embodiment, in the processor **42**, the acquisition unit **42A** acquires the ambient light information **64** that is information on the ambient light on the periphery of the exterior body. The control via the light source controller **42B** includes control of adjusting the light emission aspect of the light-emitting unit **26** based on the acquired ambient light information **64**. Accordingly, since the light emission aspect is adjusted in accordance with the ambient light on the periphery of the exterior body **12**, the light L transmitted through the exterior body **12** can be made to be close to the intended recognition state as compared to a case in which the same light emission aspect is always maintained even in a case in which the ambient light changes.

[0130] In addition, in the electronic apparatus **10** according to the second embodiment, the ambient light information **64** includes the ambient light intensity information **68**. The control via the light source controller **42B** includes control of adjusting the light emission amount of the light-emitting unit **26** based on the ambient light intensity information **68**. Accordingly, since the light emission amount is adjusted in accordance with the brightness of the periphery of the exterior body **12**, the brightness of the light L transmitted through the exterior body **12** can be made to be close to the intended brightness as compared to a case in which the light emission amount is always the same even in a case in which the brightness of the ambient light is changed.

[0131] In addition, in the electronic apparatus **10** according to the second embodiment, the ambient light information **64** includes the ambient light color information **66**. The control via the light source controller **42B** includes control of adjusting the light emission color of the light-emitting unit **26** based on the ambient light color information **66**. Accordingly, since the light emission color is adjusted in accordance with the color of the ambient light on the periphery of the exterior body **12**, the light L transmitted through the exterior body **12** can be made to be close to the intended emission color as compared to a case in which the light emission color is always the same even in a case in which the color of the ambient light is changed.

[0132] It should be noted that, in the second embodiment, although the form example is described in which the light emission aspect of the light-emitting unit **26** is adjusted based on both the ambient light color information **66** and the ambient light intensity information **68**, the present disclosed technology is not limited to this. For example, an aspect may be adopted in which only the adjustment of the light emission color of the light-emitting unit **26** is performed based on the ambient light color information **66**, or an aspect may be adopted in which only the adjustment of the light emission amount of the light-emitting unit **26** is performed based on the ambient light intensity information **68**.

Third Embodiment

[0133] In the first embodiment, the form example is described in which the exterior body color information **54** is used as the exterior body information **52** and the adjustment of the light emission aspect is performed based on the exterior body color information **54**, but the present disclosed technology is not limited to this. In the third embodiment, the exterior body information **52** includes information on a movable portion (movable mechanism) **72** of the exterior body **12**.

[0134] As illustrated in FIG. **14** as an example, the exterior body **12** comprises the movable portion **72**. The movable portion **72** is a part of the exterior body **12** and is a portion that is movable with respect to a main body of the exterior body **12**. In the example illustrated in FIG. **14**, the movable portion **72** is a cover member **72A** provided at a position facing the light-emitting unit **26**. The cover member **72A** is openable and closable, and the light-emitting unit **26** is exposed to the outside of the exterior body **12** in a case in which the cover member **72A** is in an open state. The cover member **72A** is, for example, a slide cover.

[0135] Further, the mobile printer **11** comprises a contact sensor **74**. The contact sensor **74** detects a position of the movable portion **72**. In the example illustrated in FIG. **14**, the contact sensor **74** detects whether or not the cover member **72A** is located at an open position. The contact sensor **74**

is, for example, a mechanical switch that is turned on in a case in which the cover member 72A is located at an open position and that is turned in a case in which the cover member 72A is located at a closed position. Then, the contact sensor 74 outputs information on the movable portion 72 based on the detection result as the exterior body information 52. That is, in the present embodiment, the exterior body information 52 includes information on the movable portion 72. In the example illustrated in FIG. 14, the contact sensor 74 outputs cover open/close information 76 to the processor 42. The cover open/close information 76 is information indicating an open/close state of the cover member 72A. The cover open/close information 76 is an example of “information on a state of a movable portion of the exterior body” according to the present disclosed technology. The cover member 72A is an example of a “cover member” according to the present disclosed technology.

[0136] In the processor 42, the acquisition unit 42A acquires the cover open/close information 76 as the information on the movable portion 72 from the contact sensor 74. Then, the acquisition unit 42A outputs the cover open/close information 76 to the light source controller 42B.

[0137] The light source controller 42B acquires the cover open/close information 76 from the acquisition unit 42A. In addition, the light source controller 42B executes control of adjusting the light emission aspect of the light-emitting unit 26 in accordance with the open/close state of the cover member 72A indicated by the cover open/close information 76. Specifically, the light source controller 42B executes control of setting the parameters by using the setting table 78. The aspect of the parameter control is the same as the aspect in the first embodiment described above, and thus the details thereof will be omitted. In addition, here, although the form example is described in which the parameters are set by using the setting table 78, this is merely an example, and the light emission aspect derivation expression may be used.

[0138] For example, in a case in which the cover member 72A is open, the light emission amount from the light-emitting element 27 is relatively reduced. As a result, even in a case in which the emission light L is directly visually recognized by the user, the light amount of the emission light L can be made close to an intended value.

[0139] In a case in which the light emission timing of the light-emitting unit 26 has arrived, the light source controller 42B causes the LED light sources 26A to 26C to emit light. As described above, the light amount of each of the LED light sources 26A to 26C is adjusted in accordance with the open/close state of the cover member 72A. Therefore, in a case in which the light L is recognized through the eyes E of the user, the light L is recognized in a state close to the light amount intended in advance.

[0140] As described so far, in the electronic apparatus 10 according to the third embodiment, the exterior body information 52 includes information on the state of the movable portion 72 of the exterior body 12. The control via the light source controller 42B includes control of adjusting the light emission aspect of the light-emitting unit 26 based on the information on the state of the movable portion 72. Accordingly, since the light emission aspect is adjusted in accordance with the state of the movable portion 72 of the exterior body 12, the light L emitted from the light-emitting unit 26 can be made to be close to an intended recognition state as compared to a case in which the light emission aspect is always the same even in a case in which the state of the movable portion 72 is changed.

[0141] In addition, in the electronic apparatus 10 according to the third embodiment, the movable portion 72 is the cover member 72A provided at a position facing the light-emitting unit 26 in the exterior body 12. The control via the light source controller 42B includes control of adjusting the light emission amount of the light-emitting unit 26 in accordance with the open/close state of the cover member 72A. Accordingly, since the light emission amount is adjusted in accordance with the opening and closing of the cover member 72A of the exterior body 12, the brightness of the light L emitted from the light-emitting unit 26 can be made to be close to the intended brightness as compared to a case in which the light emission amount is always the same regardless of whether

the cover member **72A** is opened or closed.

[0142] It should be noted that, in the third embodiment, the cover member **72A** is described as the movable portion **72**, but the present disclosed technology is not limited to this. The movable portion **72** may be, for example, a switch member or the like.

Fourth Embodiment

[0143] In the first embodiment, the form example is described in which the exterior body color information **54** is used as the exterior body information **52** and the adjustment of the light emission aspect is performed based on the exterior body color information **54**, but the present disclosed technology is not limited to this. In the fourth embodiment, the exterior body information **52** includes the posture information **82** that is information indicating the posture of the electronic apparatus **10**.

[0144] As illustrated in FIG. **15** as an example, the mobile printer **11** comprises a posture sensor **80**. The posture sensor **80** detects the posture of the mobile printer **11**. Examples of the posture sensor **80** include an inertial sensor and an acceleration sensor. Then, the posture sensor **80** outputs the posture information **82** as the exterior body information **52**. That is, in the present embodiment, the exterior body information **52** includes the posture information **82**. The posture information **82** is information indicating the posture of the electronic apparatus **10** (here, the mobile printer **11**). The posture of the electronic apparatus **10** includes a rotation angle about a center axis in the front-rear direction, the left-right direction, and the up-down direction from a reference position in a case in which the electronic apparatus **10** is placed on a horizontal plane. In addition, the posture of the electronic apparatus **10** includes a movement speed in each of the front-rear direction, the left-right direction, and the up-down direction of the electronic apparatus **10**. The posture information **82** is an example of “posture information” according to the present disclosed technology.

[0145] In the processor **42**, the acquisition unit **42A** acquires the posture information **82** from the posture sensor **80**. Then, the acquisition unit **42A** outputs the posture information **82** to the light source controller **42B**.

[0146] The light source controller **42B** acquires the posture information **82** from the acquisition unit **42A**. In addition, the light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** in accordance with the posture of the mobile printer **11** indicated by the posture information **82**.

[0147] Specifically, the light source controller **42B** executes the control of setting the parameters. More specifically, the light source controller **42B** acquires a light emission aspect derivation expression **84** from the storage **44**. The light emission aspect derivation expression **84** is an arithmetic expression in which a value indicating the posture (for example, the rotation angle about the XYZ axes) of the electronic apparatus **10** is an independent variable and the parameter value (for example, the duty ratio of the driving pulse or the total light emission time T) is a dependent variable. The light emission aspect derivation expression **84** is obtained, for example, by adjusting the light emission color and the light emission amount in accordance with the posture of the electronic apparatus **10** in a sensory test using an actual machine. It should be noted that, here, although the form example is described in which the parameters are set by using the light emission aspect derivation expression **84**, this is merely an example, and a derivation table may be used.

[0148] For example, in a case in which the electronic apparatus **10** is upside down, the light emission via the light-emitting unit **26** is stopped. Therefore, it is easy for the user to recognize that the posture of the electronic apparatus **10** is different from the intended posture. In addition, for example, in a case in which the electronic apparatus **10** is inclined, the light emission amount of the LED light sources **26A** to **26C** is changed in order in accordance with the inclination of the light-emitting unit **26** (for example, the light emission amount is smaller as the inclination goes to the lower side). Accordingly, the operation of the electronic apparatus **10** can be visually performed by the light emission of the light-emitting unit **26**, the user can feel fun, and the improvement in the user experience is implemented.

[0149] As described so far, in the electronic apparatus **10** according to the fourth embodiment, the exterior body information **52** includes the posture information **82** indicating the posture of the electronic apparatus **10**. The control via the light source controller **42B** includes control of adjusting the light emission aspect of the light-emitting unit **26** based on the posture information **82**. Therefore, since the light emission aspect is adjusted in accordance with the posture of the electronic apparatus **10**, the improvement in the user experience is implemented as compared to a case in which the same light emission aspect is always used even in a case in which the posture is changed.

[0150] The disclosed technologies in each of the above-described embodiments and the first modification example may be implemented in combination with each other as appropriate.

Second Modification Example

[0151] In the second modification example, in addition to the exterior body information **52** according to each of the above-described embodiments, the exterior body information **52** includes operation state information **86** indicating the operation state of the electronic apparatus **10**.

[0152] As illustrated in FIG. **16** as an example, it is assumed that the user presses a power button **19** via a finger **F1**. In such a case, the operation state information **86** is output from the power button **19** to the processor **42**. Here, the operation state information **86** is information indicating that the power of the electronic apparatus **10** (here, the mobile printer **11**) is turned on. In the present embodiment, the exterior body information **52** includes the operation state information **86**. In the processor **42**, the acquisition unit **42A** acquires the operation state information **86** of the mobile printer **11**. Then, the acquisition unit **42A** outputs the operation state information **86** to the light source controller **42B**. The operation state information **86** is an example of “operation state information” according to the present disclosed technology.

[0153] The light source controller **42B** acquires the operation state information **86** from the acquisition unit **42A**. In addition, the light source controller **42B** executes control of adjusting the light emission aspect of the light-emitting unit **26** in accordance with the operation state of the mobile printer **11** indicated by the operation state information **86**. Specifically, the light source controller **42B** executes control of setting the parameters by using the light emission aspect derivation expression **88**. The aspect of the parameter control is the same as the aspect in the fourth embodiment described above, and thus the details thereof will be omitted. It should be noted that, here, although the form example is described in which the parameters are set by using the light emission aspect derivation expression **88**, this is merely an example, and a derivation table may be used.

[0154] For example, in a case in which the power is turned on, the light-emitting unit **26** emits white light until the activation of the electronic apparatus **10** is completed. Therefore, it is easy for the user to visually recognize that the electronic apparatus **10** is being activated. By visually performing the activation of the electronic apparatus **10** by the light emission of the light-emitting unit **26**, the user can feel the fun, and the improvement in the user experience is implemented.

[0155] As described so far, in the electronic apparatus **10** according to the second modification example, the exterior body information **52** includes operation state information **86** indicating the operation state of the electronic apparatus **10**. The control via the light source controller **42B** includes control of adjusting the light emission aspect of the light-emitting unit **26** based on the operation state information **86**. Accordingly, since the light emission aspect is adjusted in accordance with the operation state of the electronic apparatus **10**, the improvement in the user experience is implemented as compared to a case in which the same light emission aspect is always maintained even in a case in which the operation state is changed.

[0156] It should be noted that, in the second modification example, the example is described in which the power is turned on as the operation state of the electronic apparatus **10**, but the present disclosed technology is not limited to this. For example, in a case in which the electronic apparatus **10** includes a battery, the remaining capacity of the battery is included in the operation state of the

electronic apparatus **10**. In such a case, the remaining capacity of the battery may be indicated by the light emission via the light-emitting unit **26**. For example, in a case in which the remaining capacity of the battery is equal to or less than the threshold value, the light-emitting unit **26** may emit red light. In addition, in a case in which the electronic apparatus **10** has the printer function, the operation state of the electronic apparatus **10** includes a progress status of the print function. In such a case, the plurality of LED light sources **26A** to **26C** may be sequentially lighted up along the discharge direction during the discharge of the printed film F. Further, the operation state of the electronic apparatus **10** includes a communication state of the electronic apparatus **10**. In such a case, the communication speed of the electronic apparatus **10** may be indicated by the light emission amount via the light-emitting unit **26**.

[0157] It should be noted that, in each of the above-described embodiments, the form example is described in which the display unit **16** is provided on the upper surface **12A** of the exterior body **12**, but the present disclosed technology is not limited to this. The display unit **16** may be provided on other surfaces of the exterior body **12**. In addition, a plurality of display units **16** may be provided on the front surface of the exterior body **12**.

[0158] Further, in each of the above-described embodiments, the form example is described in which the transmission suppression layer **30** is formed by printing, but the present disclosed technology is not limited to this. The transmission suppression layer **30** may be formed by attaching a seal member (not illustrated). Specifically, in the first member **22** of the exterior body **12**, the seal member is attached to the inner surface **22A1** of the bottom plate portion **22A**. The seal member has, for example, a laminated structure and includes a pressure-sensitive adhesive layer, the transmission suppression layer **30**, and a coating layer. The seal member is attached to the exterior body **12** via the pressure-sensitive adhesive layer. The transmission suppression layer **30** corresponding to the content shown by the display unit **16** is formed on the pressure-sensitive adhesive layer. The transmission suppression layer **30** contains the material that absorbs the wavelength in the visible light range. Further, the transmission suppression layer **30** is protected by the coating layer. The seal member is attached to form the transmission suppression layer **30** in the region corresponding to the display unit **16**. Therefore, the difference in the transmittance of the emission light L emitted from the light-emitting unit **26** is generated between the region corresponding to the display unit **16** and the region on the periphery. As a result, in a case in which the seal member is attached, the transmission suppression layer **30** can be formed, which can contribute to reduction in manufacturing cost of the electronic apparatus **10**.

[0159] In addition, in each of the above-described embodiments, the form example is described in which the three-dimensional shape of the display unit **16** protrudes rather than the periphery, but the present disclosed technology is not limited to this. For example, the three-dimensional shape of the display unit **16** may be a shape that is recessed rather than the periphery, and the periphery may be roughened and the display unit **16** may be a smooth surface.

[0160] In addition, in each of the above-described embodiments, the form example is described in which the transmission suppression layer **30** is formed by printing, but the present disclosed technology is not limited to this. For example, the transmission suppression layer **30** may be formed by being partially molded with a resin that is less likely to allow the transmission of the light, or may be formed by attaching a light shielding cover of a separate component to the inside of the exterior body **12**.

[0161] In addition, in each of the above-described embodiments, the form example is described in which the transmission of the emission light L is suppressed in the region corresponding to the display unit **16**, but the present disclosed technology is not limited to this. For example, an aspect may be adopted in which the transmission of the emission light L is promoted in the region corresponding to the display unit **16**. Specifically, the transmission suppression layer **30** may be formed in a portion other than the region corresponding to the display unit **16**, to promote the transmission of light in the region corresponding to the display unit **16**. In this way, the difference

in the light transmittance is generated between the region corresponding to the display unit **16** and the periphery.

[0162] In addition, in each of the above-described embodiments, the form example is described in which the stripe pattern having the periodic recessed and protruding portions is formed on the front surface of the exterior body **12**, but the present disclosed technology is not limited to this. The front surface of the exterior body **12** may have a structure that enhances the designability other than the stripe pattern, or may be a smooth surface.

[0163] In addition, in each of the above-described embodiments, the form example is described in which the first member **22** and the second member **24** have a symmetrical shape in a cross-sectional view, but the present disclosed technology is not limited to this. For example, the first member **22** may include the side wall portion **22B**, and the second member **24** may be a plate-like member. In such a case, an end surface of the side wall portion **22B** is in contact with the peripheral edge of the second member **24**.

[0164] In addition, in each of the above-described embodiments, the form example is described in which the exterior body **12** has a rectangular parallelepiped shape, but the present disclosed technology is not limited to this. For example, the exterior body **12** may have a square shape, a spherical shape, or a cylindrical shape.

[0165] In addition, in each of the above-described embodiments, the form example is described in which the electronic apparatus **10** is the mobile printer **11**, but the present disclosed technology is not limited to this. The electronic apparatus **10** may be, for example, a portable electronic apparatus, such as an instant camera, a digital camera, a smartphone, a tablet terminal, a laptop computer, or a mobile battery. In addition, the electronic apparatus **10** may be a stationary electronic apparatus, such as a desktop computer, a printer, a television, or a display.

[0166] In addition, in each of the above-described embodiments, the form example is described in which the film **F** is a so-called instant film containing the photosensitive material, but the present disclosed technology is not limited to this. For example, printing paper containing a heat sensitive material may be used instead of the film **F**.

[0167] In addition, in each of the above-described embodiments, the form example is described in which the light-emitting unit **26** includes the LED light source, but the present disclosed technology is not limited to this. For example, the light-emitting unit **26** may include an organic electro-luminescence (EL) element.

[0168] In addition, in each of the above-described embodiments, the form example is described in which the coating film **32** is thick as a whole in the side wall portion **22B**, but the present disclosed technology is not limited to this. The coating film **32** on half of the side wall portion **22B** on the second member **24** side may be thick.

[0169] In addition, in each of the above-described embodiments, the form example is described in which the coating film **32** is thin as a whole in the bottom plate portion **22A**, but the present disclosed technology is not limited to this. The coating film **32** may be thin in an elliptical region including the transmission portion **36**, or may be thin only in a region corresponding to the transmission portion **36**.

[0170] In addition, in each of the above-described embodiments, the form example is described in which the substrate **34** of the first member **22** is formed of the translucent material, but the present disclosed technology is not limited to this. For example, the substrate **34** may be the transparent material. However, as described above, it is preferable that the substrate **34** is formed of the translucent material from the viewpoint that the appearance adjustment by the coating film **32** is facilitated.

[0171] In addition, in each of the above-described embodiments, the form example is described in which the second member **24** is formed of the resin of the non-transparent material, but the present disclosed technology is not limited to this. For example, the second member **24** may also have a configuration in which a coating film (for example, a coating film containing titanium oxide) is

formed on a substrate of the translucent material. However, as described above, it is desirable that the second member **24** is formed of the non-transparent material from the viewpoint that the entry of the external light can be suppressed.

[0172] In addition, in each of the above-described embodiments, the form example is described in which the control program **44A** is stored in the storage **44**, but the present disclosed technology is not limited to this. For example, the control program **44A** may be stored in a storage medium (not illustrated), such as an SSD or a universal serial bus (USB) memory. The storage medium is a portable computer-readable non-transitory storage medium. The control program **44A**, which is stored in the storage medium, is installed in the electronic apparatus **10**. The processor **42** executes the light emission control processing according to the control program **44A**.

[0173] In addition, the control program **44A** may be stored in a storage device of another computer, server, or the like connected to the electronic apparatus **10** via a network, and the control program **44A** may be downloaded and installed in the electronic apparatus **10** in response to a request of the electronic apparatus **10**. That is, the program (that is, a program product) described in the present embodiment may be provided in a form of the recording medium or may be distributed from an external computer.

[0174] Moreover, in the embodiments described above, the processor **42**, the storage **44**, and the RAM **46** of the electronic apparatus **10** are described as the computer, but the present disclosed technology is not limited to this, and a device including an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or a programmable logic device (PLD) may be applied instead of the computer. Further, a hardware configuration and a software configuration may be used in combination, instead of the computer.

[0175] The following various processors can be used as hardware resources for executing the light emission control processing described in each of the above-described embodiments. Examples of the processor include a CPU that is a general-purpose processor functioning as the hardware resource for executing the processing by executing software, that is, a program. Examples of the processor also include a dedicated electronic circuit that is a processor having a dedicated circuit configuration specially designed to execute specific processing, such as an FPGA, a PLD, or an ASIC. A memory is built in or connected to any processor, and any processor executes the processing by using the memory.

[0176] The hardware resource for executing the light emission control processing may be configured by one of these various processors, or may be configured by a combination (for example, a combination of a plurality of FPGAs or a combination of a CPU and an FPGA) of two or more processors of the same type or different types. In addition, the hardware resource for executing the light emission control processing may be one processor.

[0177] As a configuring example of one processor, first, there is a form in which one processor is configured by a combination of one or more CPUs and software and the processor functions as the hardware resource for executing the light emission control processing. Second, as represented by system-on-a-chip (SoC), there is a form in which a processor that realizes the functions of the entire system including a plurality of hardware resources for executing the light emission control processing with a single integrated circuit (IC) chip is used. As described above, the light emission control processing is implemented by using one or more of the various processors as the hardware resources.

[0178] As the hardware structures of various processors, more specifically, an electronic circuit in which circuit elements, such as semiconductor elements, are combined can be used. Further, the above-described processing is merely an example. Therefore, it goes without saying that unnecessary steps may be deleted, new steps may be added, or the processing order may be changed within a range that does not deviate from the scope.

[0179] The above-described contents and the above-illustrated contents are detailed descriptions of portions related to the present disclosed technology, and are merely examples of the present

disclosed technology. For example, the description of the configuration, the function, the operation, and the effect are the description of examples of the configuration, the function, the operation, and the effect of the portion according to the present disclosed technology. Therefore, it goes without saying that unnecessary portions may be deleted, new elements may be added, or replacements may be made with respect to the above-described contents and the above-illustrated contents within a range that does not deviate from the scope of the present disclosed technology. Further, the description of, for example, common technical knowledge that does not need to be particularly described to enable the implementation of the present disclosed technology is omitted in the above-described contents and the above-illustrated contents in order to avoid confusion and to facilitate the understanding of the portions related to the present disclosed technology.

[0180] In the present specification, “A and/or B” is synonymous with “at least one of A or B”. That is, “A and/or B” may mean only A, only B, or a combination of A and B. In the present specification, the same concept as “A and/or B” also applies to a case in which three or more matters are expressed by association with “and/or”.

[0181] All of the documents, the patent applications, and the technical standards described in the present specification are incorporated into the present specification by reference to the same extent as in a case in which the individual documents, patent applications, and technical standards are specifically and individually noted to be described by reference.

[0182] In regard to the embodiment described above, the following supplementary notes will be further disclosed.

Supplementary Note 1

[0183] An electronic apparatus comprising: an exterior body; a light-emitting unit that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body; and a processor, in which the processor acquires exterior body information that is information on a state of the exterior body, and executes control of adjusting a light emission aspect of the light-emitting unit based on the acquired exterior body information.

Supplementary Note 2

[0184] The electronic apparatus according to supplementary note 1, in which the exterior body information includes exterior body color information that is information on a color of the exterior body, and the control includes control of adjusting a light emission color of the light-emitting unit based on the exterior body color information.

Supplementary Note 3

[0185] The electronic apparatus according to supplementary note 1 or 2, in which the exterior body information includes transmitted light color information that is information on a change in a color of the light transmitted through the exterior body, and the control includes control of adjusting a light emission color of the light-emitting unit based on the transmitted light color information.

Supplementary Note 4

[0186] The electronic apparatus according to any one of supplementary notes 1 to 3, in which the exterior body information includes transmitted light intensity information that is information on a change in intensity of the light transmitted through the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting unit based on the transmitted light intensity information.

Supplementary Note 5

[0187] The electronic apparatus according to any one of supplementary notes 1 to 4, in which the processor further acquires ambient light information that is information on ambient light of a periphery of the exterior body, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the acquired ambient light information.

Supplementary Note 6

[0188] The electronic apparatus according to supplementary note 5, in which the ambient light information includes ambient light intensity information indicating brightness of the periphery of

the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting unit based on the ambient light intensity information.

Supplementary Note 7

[0189] The electronic apparatus according to supplementary note 5 or 6, in which the ambient light information includes ambient light color information indicating a color of the ambient light of the periphery of the exterior body, and the control includes control of adjusting a light emission color of the light-emitting unit based on the ambient light color information.

Supplementary Note 8

[0190] The electronic apparatus according to any one of supplementary notes 1 to 7, in which the exterior body information includes information on a state of a movable portion of the exterior body, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the information on the state of the movable portion.

Supplementary Note 9

[0191] The electronic apparatus according to supplementary note 8, in which the movable portion is a cover member that is provided at a position facing the light-emitting unit in the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting unit in accordance with an open/close state of the cover member.

Supplementary Note 10

[0192] The electronic apparatus according to any one of supplementary notes 1 to 9, in which the exterior body information includes posture information indicating a posture of the electronic apparatus, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the posture information.

Supplementary Note 11

[0193] The electronic apparatus according to any one of supplementary notes 2 to 10, in which the exterior body information includes operation state information indicating an operation state of the electronic apparatus, and the control includes control of adjusting the light emission aspect of the light-emitting unit based on the operation state information.

Supplementary Note 12

[0194] The electronic apparatus according to any one of supplementary notes 1 to 11, in which a stripe pattern consisting of periodic recessed and protruding portions is formed on a front surface of the exterior body.

Supplementary Note 13

[0195] The electronic apparatus according to any one of supplementary notes 1 to 12, in which the exterior body has a display unit that shows a character, a symbol, and/or a figure by using a difference in a three-dimensional shape with a periphery on a front surface, and the display unit is formed at a position facing the light-emitting unit.

Supplementary Note 14

[0196] The electronic apparatus according to any one of supplementary notes 1 to 13, in which a coating film is formed on an outer side of the exterior body, and a thickness of the coating film in a region through which the light from the light-emitting unit is transmitted is set to be less than a thickness of the coating film in another region.

Supplementary Note 15

[0197] The electronic apparatus according to any one of supplementary notes 1 to 14, in which the light-emitting unit includes an LED light source, and the control includes control of adjusting a light emission aspect of the LED light source.

EXPLANATION OF REFERENCES

Claims

- 1.** An electronic apparatus comprising: an exterior body; a light-emitting device that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body; and a processor, wherein the processor acquires exterior body information that is information on a state of the exterior body, and executes control of adjusting a light emission aspect of the light-emitting device based on the acquired exterior body information.
- 2.** The electronic apparatus according to claim 1, wherein the exterior body information includes exterior body color information that is information on a color of the exterior body, and the control includes control of adjusting a light emission color of the light-emitting device based on the exterior body color information.
- 3.** The electronic apparatus according to claim 1, wherein the exterior body information includes transmitted light color information that is information on a change in a color of the light transmitted through the exterior body, and the control includes control of adjusting a light emission color of the light-emitting device based on the transmitted light color information.
- 4.** The electronic apparatus according to claim 1, wherein the exterior body information includes transmitted light intensity information that is information on a change in intensity of the light transmitted through the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting device based on the transmitted light intensity information.
- 5.** The electronic apparatus according to claim 1, wherein the processor further acquires ambient light information that is information on ambient light of a periphery of the exterior body, and the control includes control of adjusting the light emission aspect of the light-emitting device based on the acquired ambient light information.
- 6.** The electronic apparatus according to claim 5, wherein the ambient light information includes ambient light intensity information indicating brightness of the periphery of the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting device based on the ambient light intensity information.
- 7.** The electronic apparatus according to claim 5, wherein the ambient light information includes ambient light color information indicating a color of the ambient light of the periphery of the exterior body, and the control includes control of adjusting a light emission color of the light-emitting device based on the ambient light color information.
- 8.** The electronic apparatus according to claim 1, wherein the exterior body information includes information on a state of a movable mechanism of the exterior body, and the control includes control of adjusting the light emission aspect of the light-emitting device based on the information on the state of the movable mechanism.
- 9.** The electronic apparatus according to claim 8, wherein the movable mechanism is a cover member that is provided at a position facing the light-emitting device in the exterior body, and the control includes control of adjusting a light emission amount of the light-emitting device in accordance with an open/close state of the cover member.
- 10.** The electronic apparatus according to claim 1, wherein the exterior body information includes posture information indicating a posture of the electronic apparatus, and the control includes control of adjusting the light emission aspect of the light-emitting device based on the posture information.
- 11.** The electronic apparatus according to claim 2, wherein the exterior body information includes operation state information indicating an operation state of the electronic apparatus, and the control includes control of adjusting the light emission aspect of the light-emitting device based on the operation state information.
- 12.** The electronic apparatus according to claim 1, wherein a stripe pattern consisting of periodic recessed and protruding portions is formed on a front surface of the exterior body.
- 13.** The electronic apparatus according to claim 1, wherein the exterior body has a display that shows a character, a symbol, and/or a figure by using a difference in a three-dimensional shape

with a periphery on a front surface, and the display is formed at a position facing the light-emitting device.

14. The electronic apparatus according to claim 1, wherein a coating film is formed on an outer side of the exterior body, and a thickness of the coating film in a region through which the light from the light-emitting device is transmitted is set to be less than a thickness of the coating film in another region.

15. The electronic apparatus according to claim 1, wherein the light-emitting device includes an LED light source, and the control includes control of adjusting a light emission aspect of the LED light source.

16. A control method of an electronic apparatus including an exterior body and a light-emitting device that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body, the control method comprising: acquiring exterior body information that is information on a state of the exterior body; and executing control of adjusting a light emission aspect of the light-emitting device based on the acquired exterior body information.

17. A non-transitory computer-readable storage medium storing a program of controlling an electronic apparatus including an exterior body and a light-emitting device that is provided inside the exterior body and that is configured to emit light to an outside through the exterior body, the program executable by a computer to execute a process comprising: acquiring exterior body information that is information on a state of the exterior body; and executing control of adjusting a light emission aspect of the light-emitting device based on the acquired exterior body information.
